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# Unit 1 - Exploring the Core

## PDF Documents

### PDF Document 1: 1738064582752-Chapter 7 - unit 1 - Exploring the Core.pdf

This PDF document is included in the unit materials.

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# UNIT 1 EXPLORING THE CORE

## Topic Investigation

### EVERYDAY PROBABILITY

Probability is a fascinating area of mathematics that deals with the likelihood of different outcomes. Every day, whether we know it or not, we make decisions based on probability. In this unit, we'll explore how probability helps us predict events and make informed decisions by using the example of the simple game Rock-Paper-Scissors.


### EVERYDAY PROBABILITY

Probability isn't just for mathematicians or game players—it's everywhere! Weather forecasts predict rain based on probability. Doctors use it to predict the likelihood of health outcomes. Investors use it to gauge potential financial gains or losses. Understanding probability helps professionals in many fields make better decisions, and it can help you too, even in everyday choices.


## EXPLORING ROCK-PAPER-SCISSORS AS A MODEL

**Rock-Paper-Scissors,** a game many of us play for fun, is actually a great model for understanding basic probability concepts. Each player chooses one of three possible moves—rock, paper, or scissors. Each move can beat one of the other two moves and lose to the other, creating a perfect balance:

Rock crushes scissors but is covered by paper.

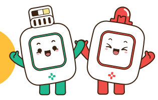
Paper covers rock but is cut by scissors.

Scissors cut paper but are crushed by rock.

## BREAKDOWN OF THE GAME

To explore the probability in this game, consider that each move has an equal chance of being played in a truly random game. Here's how the probabilities break down:

### PROBABILITY OF ANY SINGLE CHOICE (ROCK, PAPER, OR SCISSORS) BEING PLAYED:



- Since there are three options, each move has a 1 in 3 chance of being chosen, which mathematically is expressed as 1/3 or approximately 33.33%.

### PROBABILITY OF WINNING:

- If you choose rock, you will win if your opponent chooses scissors. Since the choice of scissors also has a probability of 1/3, the probability of you winning with rock is 33.33%.

### PROBABILITY OF LOSING:

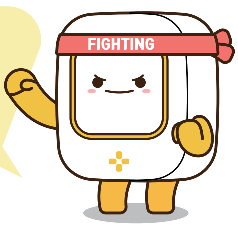
- Similarly, if you choose rock, you will lose if your opponent chooses paper, which again is 33.33%.

### PROBABILITY OF A TIE:

- If both players choose the same move, the game is a tie. The probability of both choosing rock, both choosing paper, or both choosing scissors is each (1/3) x (1/3) = 1/9. Thus, the probability of any tie occurring is (1/9) +(1/9) + (1/9) = 3/9 = 1/3, which can be expressed as 33.33%.

**TIPS**

While Rock-Paper-Scissors seems based purely on chance, skilled players use psychology and pattern recognition to predict and counter their opponents' moves, turning it into a strategic battle!



## Inquiry & Reflection

### Discussion question 1

Can you think of a way to win more games of Rock-Paper-Scissors? Consider if changing your moves often could help you win.

### Discussion question 2

How does probability show up in our daily lives? Think of an example where you use probability to make decisions every day.